The Ephemeris

December 2019
Volume 30 Number 04- The Official Publication of the San Jose Astronomical Association

December 2019 - March 2020 Events

Board & General Meetings
Saturday 12/14, 1/11, 2/8, 3/7
Board Meetings: 6 -7:30pm
General Meetings: 7:30-9:30pm

Fix-It Day (2-4pm)
Sunday 12/1, 1/5, 2/9, 3/8

Solar Sunday (1:30-3:30pm)
Sunday 12/1, 1/5, 2/9, 3/8

Intro to the Night Sky Class &
Houge Park 1Q In-Town Star Party
Friday 12/6, 1/3, 1/31

Houge Park 3Q In-Town Star Party
Friday 12/20, 1/17, 2/14, 3/13

RCDO Starry Nights Star Party
Saturday 12/21, 1/18, 2/15, 3/14

Imaging SIG Mtg
Tuesday 12/17, 1/21, 2/18, 3/17

Astro Imaging Clinics
(at Little Uvas OSP)
Saturday 12/28, 3/21

Astro Imaging Workshops
(at Little Uvas OSP)
Saturday 1/25, 2/22

Binocular Star Gazing
Saturday (will resume in Spring 2020)

Please refer to the SJAA Website home page (www.sjaa.net) and click on any "Upcoming Event" highlighted in Blue. It will take you to the SJ-Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events/) where you can find specific event times, locations, maps and possible cancellation due to weather.

INSIDE THIS ISSUE

Photo by SJAA Member: Glenn Newell
Nov 2,3 2019
Location: Union City, CA

SJAA Turns 65

Credit: Rob Jaworski

At the end of this year, 2019, the San Jose Astronomical Association is celebrating its sixty fifth birthday. For us mere mortals, sixty five is traditionally the age at which we retire, quit our decades -long journey of work and getting things done, and finally look forward to hopefully another couple of decades of rest and relaxation. I don't see the same thing for the SJAA.

The mission of the SJAA is to bring astronomy and science to the public, to inspire curiosity and to provide a place to go when an interest in space or astronomy (or geology or chemistry or any kind of science) is kindled. In these short-attention-span times, it seems that NASA missions (to the moon or Mars!) or rare celestial events (the recent Great American Eclipse of 2017!) or the sight of rocket launches (except maybe for SpaceX launches) don't have the heft to inspire the awe that they once did.

I submit that something about those events being viewed on a high definition, but really small screen, and the number of times that it's possible to view them, causes dilution of the experience, and dilution of the awe.

The handful of volunteers who run SJAA know that the awe is still there. The awe has been instilled deep within them, and they know there are seeds of that awe hibernating in others. Those seeds want to grow, but it's not easy for them to germinate. Too many distractions, too many choices on what to do. But despite all that, SJAA volunteers continue to give their time to provide that place, either in the lecture hall or out at a dark hilltop, so the seed can be provided what it needs to grow into a healthy interest and inquiry in the world around us and beyond.

In the sixty five years since a small group of neighbors got together to create this special interest group, a few years before Sputnik and well before Apollo 11, there have been thousands of kids and grown-ups (and kids that turned into grown-ups), who have availed themselves of what SJAA has to offer. They may have come to listen to a professional astronomer talk about her research into the origin of galaxies. Or they may have come to borrow a telescope from the long-running Telescope Loaner program. Or they simply wanted to bring their kids out into the local night sky and show them what Mars or Saturn, or even just the moon, looks like up close and personal. Most of those people had a fleeting, momentary experience with the SJAA. Some of them became members for a period of time before interests shifted and they moved on. Still others, like a few I can name off the top of my head, stuck around for decades as members (and are still around, as members). And some members took the next step and donated the most precious of assets, their time. They became volunteers because they loved what they were learning and wanted to share it with others.

Since the SJAA is entirely a volunteer -run organization, we are always on the lookout for more members to participate. Attending events as a member of the public (or a member of SJAA) is good and very much appreciated. Participating and helping out is even better! Whatever your skills or strengths, you can help make a difference. Come to an event, talk to any board member of someone "in charge", and they can tell you more. It can be uncomfortable at first, but you'll soon find that it's a great experience. It's what you make it.

Sixty five years is a long time for a non-profit organization to exist. Although the SJAA is not a very well known staple of the south bay, a staple it is and we are proud that a steady stream of newcomers discovers the club and works to keep it going. We are thankful to those who bring new energy to the club along with their excitement to share the heavens with others. We are thankful to the City of San Jose for providing a place to call our basecamp. We are thankful that the SJAA has been here since 1954, and we will work to keep it going for another 65 years!

With respect,

Rob Jaworski

Treasurer, past President and board member

Current Open Positions

Credit: Swami Nigam

Open SJAA Positions:

SJAA is an entirely volunteer run organization, and as such relies on the generous donation of time by its members towards running member and community events, as well as managing the club operations. Currently we have the following positions open for which we are seeking motivated individuals willing to contribute to the success of SJAA.

Volunteer Coordinator: The volunteer coordinator communicates with the group of SJAA volunteers to perform various tasks at SJAA events or programs. The volunteer coordinator will monitor membership signups/renewals to identify and communicate with new volunteers, connect volunteers with the relevant program leader or event coordinator, and be generally available for volunteers to contact in case of questions. S/he will be responsible for maintaining volunteer lists, maintaining tasks list, and matching volunteers with the tasks. S/he will also be responsible for the content on the "How to Volunteer" page of our website.

SJAA Liaison & Outreach Coordinator: SJAA receives a fair number of collaboration requests from various organizations who wish to work with SJAA to bring astronomy to their members. The SJAA Liaison & Outreach Coordinator is responsible for managing communications with all such organizations or individuals who wish to collaborate with SJAA or request SJAA involvement in their event. S/he will use the SJAA's Outreach Guidelines to determine SJAA's participation in the event, and will be responsible for managing all logistics of such participation.

Ephemeris Content Editor: The Ephemeris Content Editor is responsible for developing the content of SJAA's quarterly newsletter, The Ephemeris. He/she is responsible for identifying articles, pictures, and other relevant information to be included in the newsletter. He/she works in collaboration with the Production Editor to create each edition of the newsletter.

Introducing the new additions Gerry Joyce and Peter Melhus

Introduction of Vice -President Gerry Joyce

Above: Gerry Joyce

I have been fascinated with the night sky since I was a child. Later in life, my interest was piqued while backpacking in the Sierras - I marveled at the Milky Way and how the night sky revolved around the North Star. Fast forwarding to retirement, I now have time to pursue this lifetime interest. After a successful attempt to photograph the Milky Way over Silicon Valley from Lick Observatory, I wondered what would happen if I attached my DSLR to a telescope. To satisfy my curiosity and learn more from people with similar interests, in 2016 I joined SJAA. I began with the observers loaner program and have since moved on to imaging. I am still learning to image with my telescope, but am now capturing and processing images. I still have much to learn about Astro Photography, and look forward to continuing my education with help from the many friendly, smart and knowledgeable SJAA members who have helped me greatly along the way. A year ago I became a docent with Santa Clara Valley Open Space Authority so I could assist with opening OSA’s dark sites for SJAA members and the public. I became the Vice President of SJAA because I wanted to give back to the organization and be more involved in acquainting the public with SJAA and its many functions.

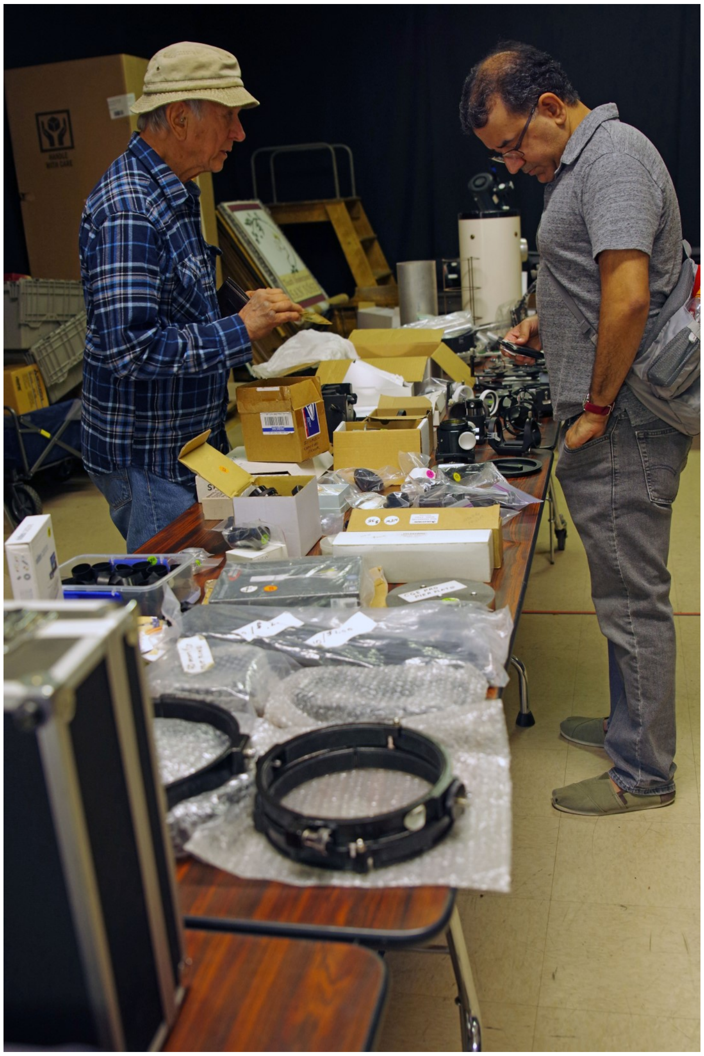
Introduction of Board Member Peter Melhus

Above: Peter Melhus

Peter brings to the Board experiences outside the astronomy field. He has been a SJAA member since March of 2018. He has 25 years of experience in the private sector, 12 years in academia and 25 years on various nonprofit boards of directors, many of which were smaller environmental groups. He's had a lifetime interest in amateur astronomy but has never had the time to pursue it until his retirement. While providing the board with governance guidance he hopes to develop his own astronomy skills. Peter holds an undergraduate degree in mechanical engineering, an MBA and a PhD in city and regional planning.

Fall Swap Meet

Credits Text: Rob Jaworski Images: Ken Miura



SJAA held its annual Fall Swap Meet on Sunday, Oct. 13 2018 at the Houge Park Hall.

The SJAA received just over $500 in donations as part of the swap meet.

The next Swap Meet is tentatively scheduled for March 29, 2020. Please watch the SJAA website for additional information as the date gets closer.

SJAA wishes its members Happy Holidays and Best Wishes for a Happy New Year 2020!

Transit Of Mercury Across the Sun

Credits Images Vini Carter and Ray Ilya

SJAA held an event on the morning of Monday, November 11, to view the transit of Mercury across the disk of the Sun. This transit is like a tiny solar eclipse, where Mercury blocks part of the Sun from our perspective, and we see Mercury move across the disk of the Sun over the course of a few hours. We had a tremendous turnout for this event. There were 15-20 solar telescopes and binoculars setup at the Houge Park telescope row, and approximately 200 people who came to view the transit while they enjoyed coffee and donuts and bagels provided by SJAA. Being a Veterans Day school holiday, there were many children also among the audience.

The next Mercury transit is scheduled for November 13, 2032. However, it will not be viewable in North America. The next Mercury transit visible in North America will not take place until 2049

Voyager 2 Illuminates Boundary of Interstellar Space

Credit: NASA JPL

One year ago, on Nov. 5, 2018, NASA's Voyager 2 became only the second spacecraft in history to leave the heliosphere - the protective bubble of particles and magnetic fields created by our Sun. At a distance of about 11 billion miles (18 billion kilometers) from Earth - well beyond the orbit of Pluto - Voyager 2 had entered interstellar space, or the region between stars. Today, five new research papers in the journal Nature Astronomy describe what scientists observed during and since Voyager 2's historic crossing.

Each paper details the findings from one of Voyager 2's five operating science instruments: a magnetic field sensor, two instruments to detect energetic particles in different energy ranges and two instruments for studying plasma (a gas composed of charged particles). Taken together, the findings help paint a picture of this cosmic shoreline, where the environment created by our Sun ends and the vast ocean of interstellar space begins.

The Sun's heliosphere is like a ship sailing through interstellar space. Both the heliosphere and interstellar space are filled with plasma, a gas that has had some of its atoms stripped of their electrons. The plasma inside the heliosphere is hot and sparse, while the plasma in interstellar space is colder and denser. The space between stars also contains cosmic rays, or particles accelerated by exploding stars. Voyager 1 discovered that the heliosphere protects Earth and the other planets from more than 70% of that radiation.

When Voyager 2 exited the heliosphere last year, scientists announced that its two energetic particle detectors noticed dramatic changes: The rate of heliospheric particles detected by the instruments plummeted, while the rate of cosmic rays (which typically have higher energies than the heliospheric particles) increased dramatically and remained high. The changes confirmed that the probe had entered a new region of space.

Before Voyager 1 reached the edge of the heliosphere in 2012, scientists didn't know exactly how far this boundary was from the Sun. The two probes exited the heliosphere at different locations and also at different times in the constantly repeating, approximately 11-year solar cycle, over the course of which the Sun goes through a period of high and low activity. Scientists expected that the edge of the heliosphere, called the heliopause, can move as the Sun's activity changes, sort of like a lung expanding and contracting with breath. This was consistent with the fact that the two probes encountered the heliopause at different distances from the Sun.

The new papers now confirm that Voyager 2 is not yet in undisturbed interstellar space: Like its twin, Voyager 1, Voyager 2 appears to be in a perturbed transitional region just beyond the heliosphere.

"The Voyager probes are showing us how our Sun interacts with the stuff that fills most of the space between stars in the Milky Way galaxy," said Ed Stone, project scientist for Voyager and a professor of physics at Caltech. "Without this new data from Voyager 2, we wouldn't know if what we were seeing with Voyager 1 was characteristic of the entire heliosphere or specific just to the location and time when it crossed."

Pushing Through Plasma

The two Voyager spacecraft have now confirmed that the plasma in local interstellar space is significantly denser than the plasma inside the heliosphere, as scientists expected. Voyager 2 has now also measured the temperature of the plasma in nearby interstellar space and confirmed it is colder than the plasma inside the heliosphere.

In 2012, Voyager 1 observed a slightly higher-than-expected plasma density just outside the heliosphere, indicating that the plasma is being somewhat compressed. Voyager 2 observed that the plasma outside the heliosphere is slightly warmer than expected, which could also indicate it is being compressed. (The plasma outside is still colder than the plasma inside.) Voyager 2 also observed a slight increase in plasma density just before it exited the heliosphere, indicating that the plasma is compressed around the inside edge of the bubble. But scientists don't yet fully understand what is causing the compression on either side.

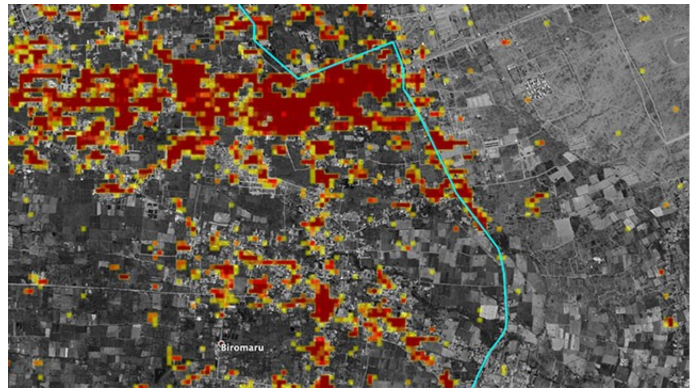
Leaking Particles

If the heliosphere is like a ship sailing through interstellar space, it appears the hull is somewhat leaky. One of Voyager's particle instruments showed that a trickle of particles from inside the heliosphere is slipping through the boundary and into interstellar space. Voyager 1 exited close to the very "front" of the heliosphere, relative to the bubble's movement through space. Voyager 2, on the other hand, is located closer to the flank, and this region appears to be more porous than the region where Voyager 1 is located.

NASA Map reveals a landslide risk factor

Credit: NASA JPL

A man retrieving coconuts from a home destroyed by landslides, with damaged rice paddies in the background. Credit: European Union/Pierre Prakash, CC BY-NC-ND 2.0



JPL's ARIA team produced this Damage Proxy Map of Palu after the 2018 earthquake. The aqueduct is marked in blue; red and yellow pixels show likely damage. The terrain slopes uphill from left to right, with major damage below the aqueduct and little above it. Credit: NASA/JPL-Caltech/JAXA

In the deadly 2018 earthquake in the Indonesian city of Palu, intense shaking changed solid ground into a landslide of flowing mud, multiplying the death toll and economic impact. A new paper shows that this disastrous effect was triggered by a previously unknown risk factor: flooding rice fields for farming.

Soil liquefaction, which causes this kind of landslide, occurs when the shaking from a large earthquake rips through moist, loose soil, overpowering the friction that normally holds dirt particles together. The soil loses its structural integrity and begins to flow like a liquid. Buildings fall as their support washes away. Heavy objects like cars sink into the muck, while buried water and sewer pipes rise to the surface.

In Palu, although early reporting blamed most of the estimated 2,000 fatalities on a tsunami, surveys soon showed that soil-liquefaction landslides caused at least as much damage as the ocean waves did. That puzzled researchers. Soil liquefaction usually occurs in flat landscapes with wet, sandy or silty ground, such as coastal plains. Researchers had thought flat terrain was a prerequisite because the water table - the distance below ground where the soil becomes saturated with water - must be shallow, and that's rare on a hillside. Palu has sandy soil, but it's in a gently sloping valley that appeared to pose little risk.

They were startled to notice that all the landslides originated along a distinct line. When they took a closer look, they saw that the line was an aqueduct. "So we started studying why the aqueduct is clearly defining the boundary between land sliding and no sliding," Yun said.

The Gumbasa Aqueduct was completed in 1913 to reduce the risk of famine by providing a consistent water supply for local farmers. Only land downhill from the aqueduct is irrigated; water is not pumped uphill. Farmers just below the aqueduct practice wet rice cultivation, in which fields are flooded at one point in the growing cycle. This predominant method of rice farming in tropical Asia raises the water table over time to just below the ground surface. Farther downhill, farmers grow coconut palms, which require less irrigation and don't raise the water table as much.

Damage maps revealed widespread liquefaction below the aqueduct. Multiple slides carried some 6 square miles (16 square kilometers) of land far downhill - in some places farther than 49 feet (15 meters). The slides were slowed or halted by the coconut palm plantations. No liquefaction was identified uphill of the aqueduct.

The conclusion is clear, Bradley said: "If there hadn't been intensive irrigation, the landslides wouldn't have happened." On the positive side, he added, "This is a human-caused hazard, and it can have a human solution. We can't mitigate the hazard of ground shaking on Palu, but the agricultural practices can be updated based on this new understanding."

Trees played a critical role in stopping the slides, and the researchers suggest that planting more trees - perhaps interspersed with rice fields - in areas that are intensively irrigated might reduce the risk of soil liquefaction.

SJAA Member Astrophoto Gallery

Date: July 28, 2019

Location: Pinnacles East, Paicines, CA

Imaging Telescope: EF24mm f/1.4L II lens

Imaging Camera: Canon EOS 5D Mark IV

Mount: Tripod

Software: Lightroom

SJAA Member Astrophoto Gallery

Dates: October 31 - November 5, 2019

Location: Palo Alto, CA

Telescope: Explore Scientific ED102 FCD100

Camera: ZWO ASI1600

Mount: iOptron CEM60

Software: Pixinsight, PHD2, Sequence Generator Pro, LightRoom, Photoshop

SJAA Member Astrophoto Gallery

Date: October 26, 2019

Location: Little Uvas Open Space Preserve, Morgan Hill

Telescope: Takahashi FS 60CB Takahashi fs60cb

Camera: Cannon T5i

Mount: Orion Sirius EQ-G Mount

Software: AstroPixelProcessor, Adobe Lightroom CC Adobe Lightroom CC, BackyardEOS

SJAA Member Astrophoto Gallery

Date: October 25, 2019

Location: Pinnacles East, Paicines, CA

Telescope: Stellarvue 9x50 Guide Scope

Camera: Lacerta M-Gen II autoguider

Mount: iOptron iEQ45 Pro

Software: Pleiades Astrophoto PixInsight 1.8 Ripely

Kid Spot Jokes:

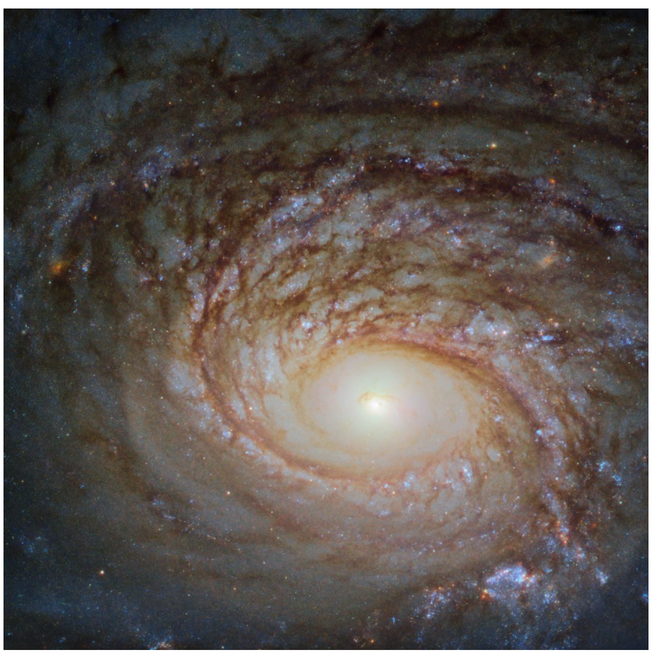
- **What did the Saturn say to its rings when they started squeezing it really hard?**
  It's a little Tit-an here.

- **What happens when he sun sneezes?**
  It's a solar storm.

- **What did the comet say to the large gaseous planet orbiting near its stars when they went on a date?**
  It's very hot Jupiter.

Hubble Spots a Curious Spiral

Credit: NASA.gov and ESA



ESA/Hubble & NASA, A. Seth et al.

The universe is simply so vast that it can be difficult to maintain a sense of scale. Many galaxies we see through telescopes such as the NASA/ESA Hubble Space Telescope, the source of this beautiful image, look relatively similar: spiraling arms, a glowing center, and a mixture of bright specks of star formation and dark ripples of cosmic dust weaving throughout.

This galaxy, a spiral galaxy named NGC 772, is no exception. It actually has much in common with our home galaxy, the Milky Way. Each boasts a few satellite galaxies, small galaxies that closely orbit and are gravitationally bound to their parent galaxies. One of NGC 772's spiral arms has been distorted and disrupted by one of these satellites (NGC 770 — not visible in the image here), leaving it elongated and asymmetrical.

However, the two are also different in a few key ways. For one, NGC 772 is both a peculiar and an unbarred spiral galaxy; respectively, this means that it is somewhat odd in size, shape or composition, and that it lacks a central feature known as a bar, which we see in many galaxies throughout the cosmos — including the Milky Way. These bars are built of gas and stars, and are thought to funnel and transport material through the galactic core, possibly fueling and igniting various processes such as star formation.

From the Board of Directors

Announcements

- ◊ Rob Chapman has stepped down from his position as Director of SJAA
- ◊ Peter Melhus joined the Board of Directors as Director 7, for the remainder of the Director term until February 2021
- ◊ Gerry Joyce was appointed as Vice President of SJAA for the remainder of the Officers term for 2019-2020
- ◊ Vikas Kache would like to step down from the position of the Content Editor for the Ephemeris due to relocation.
- ◊ SJAA would like to fill 3 open positions: Volunteer Coordinator, SJAA Liaison and Outreach Coordinator, and Ephemeris Content Editor. Please see the descriptions of these positions on Page 3 in the newsletter.
- ◊ New board member elections will be held at the February 2020 board meeting on February 8. The following 4 board positions will be open for Elections: Director 2, Director 4, Director 6, and Director 8.
- ◊ New officer elections will be held at the March 2020 board meeting on March 7.
- ◊ SJAA Observers group is being rehosted from Yahoo to Google
- ◊ New Google group for SJAA members is being created as a communications channel to reach all SJAA members in good standing.
- ◊ Next Swap Meet tentatively scheduled for March 29, 2020

Board Meeting Excerpts

September 14, 2019

In attendance
Swami, Sukhada, Glenn, Vini, Rob J, Wolf
Excused Absences: Rob C, Ken, Sandy
Guests: Peter, Gerry, Bob and Natti, Brendan, Dimitry

Rob C stepped down
Rob Chapman indicated to the board that he would like to step down from his position as Director 7 of SJAA. The board accepted his resignation and opened the seat for a new Director.

2020 Events Calendar
The 2020 events and operations calendar is in the process of being finalized..

Solar Program
Money was approved for the Solar Viewing program to acquire additional equipment, such as iOptron tripod, a couple of chairs, extra daytime monitor, etc.

Outreach Events
SJAA participated in 2 outreach events - The City of Cupertino's Astronomy Nights on August 17, and the Cambrian Festival in San Jose on August 25. Both events were successful and very well received.

Website Migration
SJAA website was migrated to a new hosting provider. Migration is complete and the website is functioning properly.

October 12, 2019

In attendance
Swami, Vini, Glenn, Sukhada, Wolf , Vini, Rob J, Ken
Excused Absences: Sandy
Guests: Peter, Gerry, Kaushal, Emi, Mark, Bruce

Peter Melhus as Director 7
Peter Melhus expressed his interest in joining the board as Director 7. He was appointed as Director 7 by unanimous decision. For a brief introduction and a picture, please see Page 4 of the newsletter.

Gerry Joyce as Vice President
Gerry Joyce expressed his interest in serving as the Vice President of SJAA. He was appointed to the position of Vice President with unanimous support from the board. For a brief introduction and a picture please see Page 4 of the newsletter.

Past Outreach Events
SJAA participated in the San Jose Harvest Festival at Martial Cottle Park in San Jose on October 5. The event was very successful.

Volunteer Requests
SJAA has received requests from some high school students who would like to volunteer for the club. We are preparing a list of projects that these students can help out with, accompanied by their parents or a responsible adult legal guardian.

Detailed Job Descriptions
SJAA is putting detailed job descriptions for the various club administrative positions and program leader positions.

Observers Yahoo Group
SJAA Observers group will be moving from a Yahoo group to a Google group.

New Google Group
A new SJAA members Group will be created as a communication channel for reaching all SJAA members in good standing.

Telescope Seminar
Wolf made a proposal to design and conduct mini-seminars on telescopes. This will work hand-in-hand with the QuickSTARt program which is expected to restart in 2020.

November 9, 2019

In attendance
Swami, Wolf, Vini, Rob J, Glenn, Sandy, Ken, Sukhada
Excused Absences: Peter M, Jerry G
Guests: Emi, Marianne

Fall Swap Meet
SJAA held the Fall Swap Meet on October 13. For images and a small summary please take a look at Page 5 from this issue.

OSA docent training
OSA is asking all docents to attend volunteer training and to repeat fingerprinting and background checks.

ITSP Program Leader
Muditha Kanchana will take over the newly created position of ITSP program leader later this year.

Past Outreach Event
SJAA participated in an outreach event at Wilcox High School. Viewing conditions were less than satisfactory due to the smoke from NorCal wildfires.

SJAA Groups
Plans to switch SJAA observers group from Yahoo to Google and to create a new SJAA Members group are in progress.

Pre-school lights
SJAA will be meeting with pre-school next to Houge Park parking lot to discuss alternative ways of turning out the lights instead of installing light covers at every ITSP.

Mercury Transit
SJAA will hold an event to view the Mercury transit of the Sun on the morning of Monday, November 11. Coffee and donuts will be provided. The event went well and a there was a lot of participants. To view images from this event please look at Page 6 of this issue.

Youth Shakespeare group
Youth Shakespeare group will be using the Houge Park hall extensively from December 1 to December 14 for their annual performance. No disruption of any SJAA event is expected during this time.

Kids Spot
The Ephemeris editors would like to know if there is sufficient interest in the Kids Spot page in the newsletter among the newsletter readers.

SJAA Library

SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

http://www.sjaa.net/programs/fixit/

Solar Observing

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

http://www.meetup.com/SJ-Astronomy/

Quick STARt Program

The Quick STARt Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick STARt program is undergoing a redesign to make it more accessible to SJAA members, so signups are currently on hold.

http://www.sjaa.net/programs/quick-start/

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, run by David Grover, each with a similar format, with only the content changing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

http://www.sjaa.net/programs/beginners-astronomy/

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met.

If you are an SJAA member and an experienced observer or have been through the SJAA Quick STARt program please fill this form to request a particular item. Please also consider donating unused equipment.

http://www.sjaa.net/programs/loaner-telescope-program/

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Bruce Braunstein, the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

http://www.sjaa.net/programs/imaging-sig/

Astro Imaging Workshops and Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astrophotography, to have a chance of seeing what it is all about.

Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's (Program Coordinator) for additional information.

http://www.sjaa.net/programs/school-star-party/

SJAA Contacts

President/Dir:	Swami Nigam
Vice President:	Gerry Joyce
Treasurer/Dir:	Rob Jaworski
Secretary/Dir:	Sandy Mohan
Director:	Peter Melhus
Director:	Vini Carter
Director:	Ken Miura
Director:	Glenn Newell
Director:	Wolf Witt
Director:	Sukhada Palav
Ephemeris Newsletter -	
Content Editor:	Vikas Kache
Prod. Editor:	Emi Nikolov
Binocular Stargazing:	Ed Wong
Fix-it Program:	Vini Carter
Hands on Imaging:	Glenn Newell
Imaging SIG:	Bruce Braunstein
Intro to the Night Sky:	David Grover
Library:	Jai Purandare
Loaner Program:	Muditha Kanchana
Memberships:	Kaushal Gangakhedkar
Publicity:	Rob Jaworski
Quick STARt	Dave Ittner
Donations:	Dave Ittner*
Solar:	Wolf Witt
Starry Nights:	Carl Wong
School Events:	Cat Cvengros
Refreshments:	Marianne Damon
Social Media Manager:	Jessica Johnson
Speakers:	Sukhada Palav
Webmaster:	Satish Vellanki

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